

Part 1. (30 points) Multiple-choice questions

Question 1: The first ionization energies of the elements _____ as you go from the bottom to the top of a group in the periodic table.

- A. decrease
B. does not change
C. increase
D. are completely unpredictable

Question 2: $[\text{Ar}]4s^2 3d^{10} 4p^3$ is the electron configuration of a(n) _____ atom.

- A. V
B. P
C. Sn
D. As

Question 3: Which of the following correctly represents the second ionization of aluminum?

- A. $\text{Al}^+ (\text{g}) + \text{e}^- \longrightarrow \text{Al}^{2+} (\text{g})$
B. $\text{Al} (\text{g}) \longrightarrow \text{Al}^+ (\text{g}) + \text{e}^-$
C. $\text{Al}^+ (\text{g}) \longrightarrow \text{Al}^{2+} (\text{g}) + \text{e}^-$
D. $\text{Al} (\text{g}) + \text{e}^- \longrightarrow \text{Al}^- (\text{g})$

Question 4: There are _____ orbitals in the second shell.

- A. 4
B. 1
C. 2
D. 8

Question 5: A compound is composed of only C, H, and O. The combustion of a 0.519-g sample of the compound yields 1.24 g of CO_2 and 0.255 g of H_2O . What is the empirical formula of the compound?

- A. $\text{C}_3\text{H}_3\text{O}$
B. CH_3O
C. $\text{C}_6\text{H}_6\text{O}$
D. $\text{C}_2\text{H}_6\text{O}_5$

Question 6: There are _____ unpaired electrons in the Lewis symbol for an oxygen atom.

- A. 4
B. 3
C. 1
D. 2

Question 7: Chlorine is much more apt to exist as an anion than sodium. This is because _____.

- A. chlorine is less metallic than sodium
B. chlorine has a greater electron affinity than sodium does
C. chlorine is bigger than sodium
D. chlorine has greater ionization energy than sodium does

Question 8: Of the following, which gives the correct order for atomic radius for Mg, Na, P, Si and Ar?

- A. $\text{Mg} > \text{Na} > \text{P} > \text{Si} > \text{Ar}$
B. $\text{Ar} > \text{Si} > \text{P} > \text{Na} > \text{Mg}$
C. $\text{Ar} > \text{P} > \text{Si} > \text{Mg} > \text{Na}$
D. $\text{Na} > \text{Mg} > \text{Si} > \text{P} > \text{Ar}$

Question 9: The valence shell of the element X contains 2 electrons in a 5s subshell. Below that shell, element X has a partially filled 4d subshell. What type of element is X?

- A. halogen
B. nonmetal
C. transition metal
D. main group element

Question 10: Which periodic table group contains only nonmetals _____?

- A. 7A
B. 6A
C. 8A
D. 5A

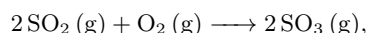
Question 11: An aqueous ethanol solution (400 mL) was diluted to 4.00 L, giving a concentration of 0.0400 M. The concentration of the original solution was _____ M.

- A. 0.400
B. 1.60
C. 2.00
D. 0.200

Question 12: Calculate ΔH° (in kJ) for reaction 3.



The enthalpy of the reaction in which sulfur dioxide is oxidized to sulfur trioxide,



is _____ kJ.

- A. 1087
B. -1384
C. 196
D. -196

Question 13: The Lewis structure of N_2H_2 shows _____.

- A. each hydrogen has one nonbonding electron pair
B. each nitrogen has one nonbonding electron pair
C. a nitrogen-nitrogen triple bond
D. each nitrogen has two nonbonding electron pairs

Question 14: Which is the correct ground-state electron configuration for silver _____?

- A. $[\text{Kr}]5s^2 4d^{10}$
B. $[\text{Kr}]5s^2 4d^9$
C. $[\text{Kr}]5s^1 4d^{10}$
D. $[\text{Xe}]5s^1 4d^{10}$

Question 15: Which electron configuration represents a violation of the Pauli exclusion principle?

- A. $\begin{array}{ccc} 1s & 2s & 2p \\ \boxed{\uparrow\downarrow} & \boxed{\uparrow} & \boxed{} \boxed{} \boxed{} \end{array}$
B. $\begin{array}{ccc} 1s & 2s & 2p \\ \boxed{\uparrow\downarrow} & \boxed{\uparrow\downarrow} & \boxed{\uparrow} \boxed{} \boxed{} \end{array}$
C. $\begin{array}{ccc} 1s & 2s & 2p \\ \boxed{\uparrow\downarrow} & \boxed{\uparrow\downarrow} & \boxed{\uparrow\uparrow} \boxed{\uparrow} \boxed{} \end{array}$
D. $\begin{array}{ccc} 1s & 2s & 2p \\ \boxed{\uparrow\downarrow} & \boxed{\uparrow\downarrow} & \boxed{\uparrow} \boxed{\uparrow\downarrow} \boxed{} \end{array}$

Question 16: Using the VSEPR model, the molecular geometry of the central atom in XeF_2 is _____.

- A. trigonal pyramidal
B. trigonal planar
C. bent
D. linear

Question 17: Oxides of the active metals combine with water to form _____.

- A. metal hydroxides B. metal hydrides C. hydrogen gas D. water and a salt

Question 18: Vanadium has two naturally occurring isotopes, ^{50}V with an atomic mass of 49.9472 amu and ^{51}V with an atomic mass of 50.9440. The atomic weight of vanadium is 50.9415. The percent abundances of the vanadium isotopes are _____% ^{50}V and _____% ^{51}V .

- A. 99.000, 1.000 B. 49.00, 51.00 C. 0.2500, 99.750 D. 1.000, 99.000

Question 19: Of the following, _____ cannot accommodate more than an octet of electrons.

- A. S B. I C. P D. O

Question 20: The ground state electron configuration of Fe is

- A. $1s^2 2s^2 2p^6 3s^2 3p^6 3d^6 4s^2$ B. $1s^2 2s^2 3s^2 3p^{10}$
C. $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2$ D. $1s^2 2s^2 3s^2 3p^6 3d^6$

Question 21: An electron in a(n) _____ subshell experiences the greatest effective nuclear charge in a many-electron atom.

- A. 3d B. 3p C. 3f D. 3s

Question 22: Given the electronegativities below, which covalent single bond is most polar?

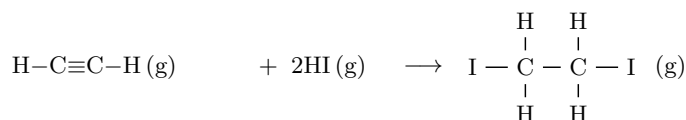
H (2.1), C (2.5), N (3.0), O (3.5)

- A. O – N B. O – H C. C – H D. N – H

Question 23: For which one of the following reactions is $\Delta H_{\text{rxn}}^\circ$ equal to the heat of formation of the product?

- A. $\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \longrightarrow 2\text{NH}_3(\text{g})$
B. $6\text{C}(\text{s}) + 6\text{H}(\text{g}) \longrightarrow \text{C}_6\text{H}_6(\text{l})$
C. $(1/2)\text{N}_2(\text{g}) + \text{O}_2(\text{g}) \longrightarrow \text{NO}_2(\text{g})$
D. $12\text{C}(\text{g}) + 11\text{H}_2(\text{g}) + 11\text{O}(\text{g}) \longrightarrow \text{C}_6\text{H}_{22}\text{O}_{11}(\text{g})$

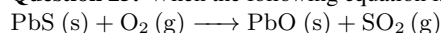
Question 24: Using the table of average bond energies below, the ΔH for the reaction is _____ kJ.



Bond:	C≡C	C – C	H – I	C – I	C – H
D (kJ/mol):	839	348	299	240	413

- A. –217 B. –160 C. +63 D. +160

Question 25: When the following equation is balanced, the coefficient of sulfur dioxide is _____.



- A. 1 B. 4 C. 2 D. 3

Question 26: Using the VSEPR model, the electron-domain geometry of the central atom in BF_3 is _____.

- A. trigonal planar B. octahedral
C. trigonal bipyramidal D. linear

Question 27: The hybrid orbital set used by the central atom in NO_3^- is _____.

- A. sp^2 B. sp^3 C. sp^3d^2 D. sp^3d

Question 28: Which one of the following polyatomic ions has the same charge as the hydroxide ion?

- A. nitrate B. carbonate C. ammonium D. sulfate

Question 29: A 50.0-g sample of liquid water at 25.0°C is mixed with 29.0 g of water at 45.0°C . The final temperature of the water is _____ $^\circ\text{C}$. The specific heat capacity of liquid water is 4.18 J/g-K.

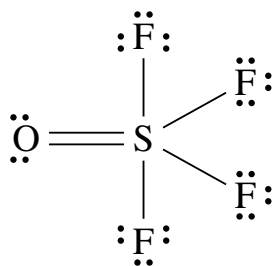
- A. 32.3 B. 102 C. 35.0 D. 27.6

Question 30: Of the following elements, which has the largest first ionization energy?

- A. Sr B. Ba C. Rb D. Ca

Part 2. (10 points) Short-answer questions

Question 31: Below is a possible Lewis structure of SOF_4 .



- (a) Prove that this structure is plausible
(b) Predict the approximate values of bond angles in this structure
(c) Use box diagrams to present the hybridization of S and indicate electrons that form σ and π bonds.

End